APPLIED AI · LECTURE



Generative AI & Large Language Models

An overview

Huseyin Kusetogullari

About the Lecture

Part	What we'll do
Welcome & warm-up	Quick poll: where have you met GenAI?
What is generative AI?	Big picture, key ideas, short history
How LLMs work	Next-word prediction, attention, training + 8-min video
Images, use & risks	Diffusion models, ways to use LLMs, limitations
Case study	GenAI for breast cancer pathology images
Wrap-up	Takeaways and discussion
About the Lecture	Overview and Final Exam

HANDS UP

Which of these are you using?

Chat assistant

Asked ChatGPT, Claude or Gemini a question

Writing help

Autocomplete or “rewrite” in email and documents

Translation

Translated a message, article or menu

Images

Generated or edited a picture with AI

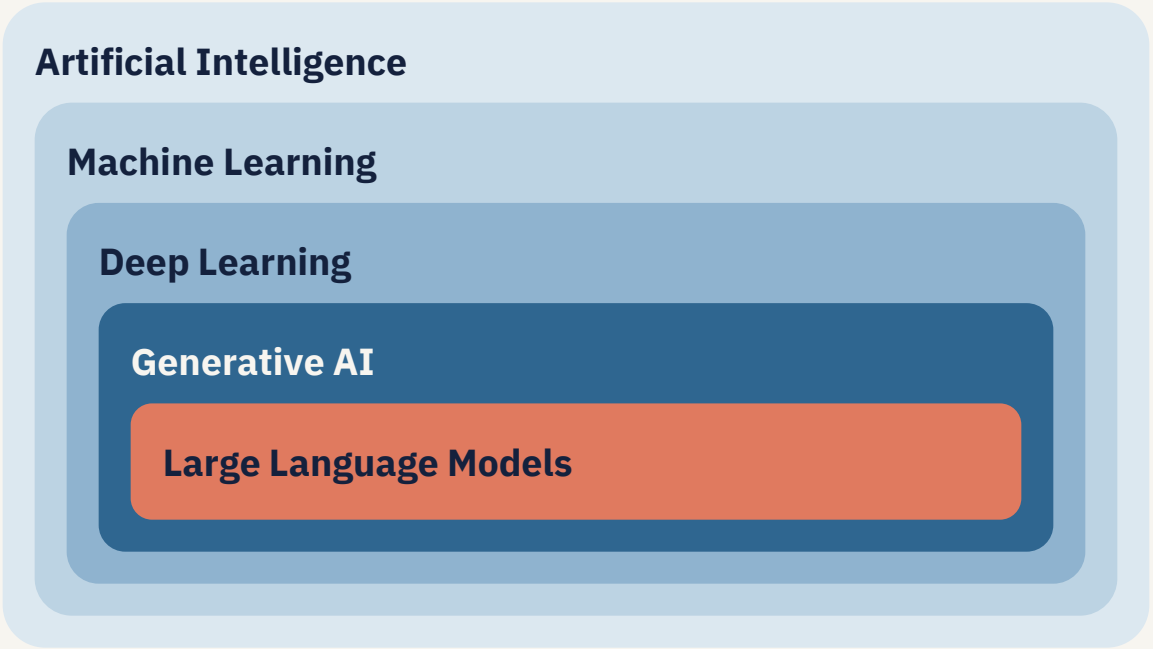
Transcripts

Auto-captions or AI meeting notes

All of these are generative AI. Today: what it is, how it works — and how it can help analyse medical images.

THE BIG PICTURE

Where GenAI and LLMs fit



- **AI** — machines doing tasks that seem to need intelligence
- **ML** — learns patterns from data instead of hand-written rules
- **Deep learning** — ML with large, many-layered neural networks
- **Generative AI** — creates new content: text, images, audio, code
- **LLMs** — generative models for language, trained on huge amounts of text

KEY IDEA

Deciding vs. creating

Discriminative AI

Learns to tell categories apart. Answers “which one?”



Output: a label

Malignant · 92%

Generative AI

Learns what the data looks like — and makes new data.

“Explain this biopsy report in plain words.”

Output: new content

“The sample shows cells that are growing abnormally…”

Same deep-learning building blocks — a different goal: **decide** vs. **create**.

WHAT IT CAN MAKE

One idea, many kinds of output

Text

Chat, summaries, translation, emails — ChatGPT, Claude, Gemini

Images

Pictures from a text prompt — Stable Diffusion, DALL·E, Midjourney

Code

Writing and fixing software — coding assistants

Audio & speech

Natural voices, music, live transcription

Video

Short clips generated from text or an image

Science & medicine

Protein structures, drug candidates, synthetic medical images

A SHORT HISTORY

Ten years from fake faces to AI agents

2014

GANs

Two networks compete — one fakes, one judges. First realistic generated faces.

2017

Transformer

“Attention Is All You Need” — the architecture behind every LLM.

2020

GPT-3

175 billion parameters. Fluent text from a short prompt.

2022

Diffusion & ChatGPT

Image generation goes public; ChatGPT reaches millions in weeks.

2023

Multimodal

GPT-4, Claude, Gemini, Llama — models that read images as well as text.

2024–26

Reasoning & agents

Models that work step by step and use tools to finish tasks.

The 2017 Transformer is the turning point: it made it practical to train on **far more data**.

THE METHODS

Five families of generative models

Zoom in next

GANs

A generator and a critic compete

Good at: sharp images, image-to-image translation

StyleGAN, CycleGAN

2013

VAEs

Compress to a short code, then rebuild

Good at: smooth variations, spotting anomalies

Inside Stable Diffusion

2020

Diffusion

Learn to remove noise, step by step

Good at: top-quality images and video

Stable Diffusion, DALL·E, Imagen

2018

Autoregressive transformers

Predict the next token, repeat

Good at: text, code, also audio

GPT, Claude, Gemini, Llama

2021

Vision–language models

Join images and text in one space

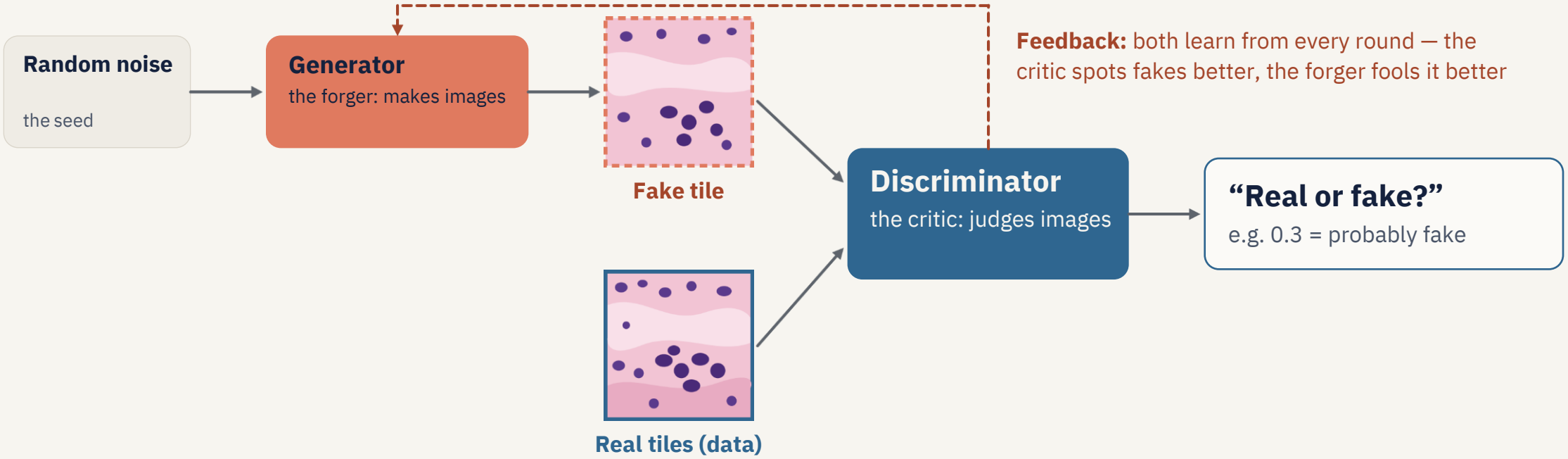
Good at: image search, captions, chat about images

CLIP, CONCH, PathChat

Real systems combine them: an image generator uses a **transformer** to read the prompt, a **VAE** to compress and **diffusion** to paint.

HOW IT WORKS · GAN

A forger and an art critic

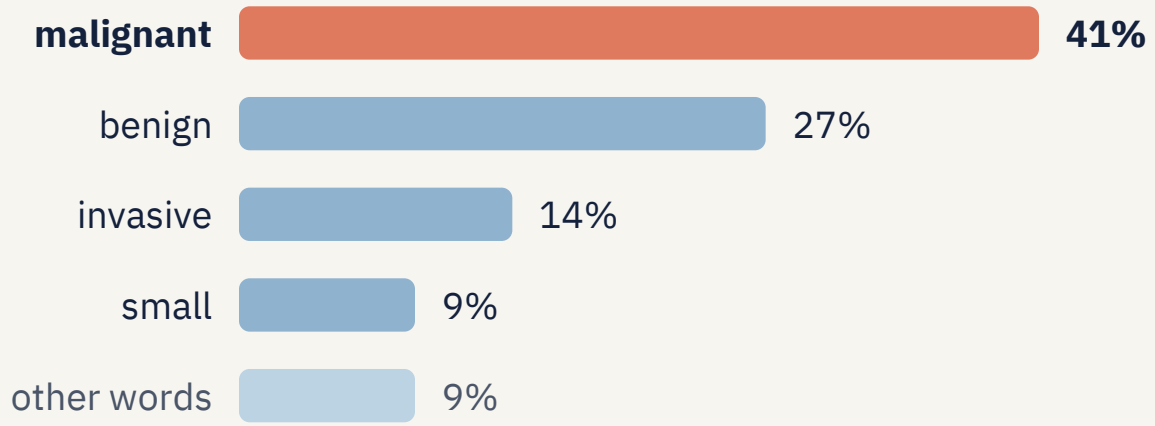


Training ends when the critic can no longer tell fake from real. **In pathology:** synthetic tiles for rare subtypes, and stain normalisation between labs (CycleGAN).

HOW LLMS WORK · 1

An LLM predicts the next word

"The biopsy showed that the tumour was …"



Probabilities are illustrative.

Writing a whole answer

1. Score every possible next word
2. Pick one — with a little randomness ("temperature")
3. Add it to the text and repeat

That's the whole trick — done with **billions of parameters**, trained on **trillions of words**.

From words to numbers

1 · Split text into tokens

Large	language	models	read	tok	ens
27433	4221	4211	1373	11241	729

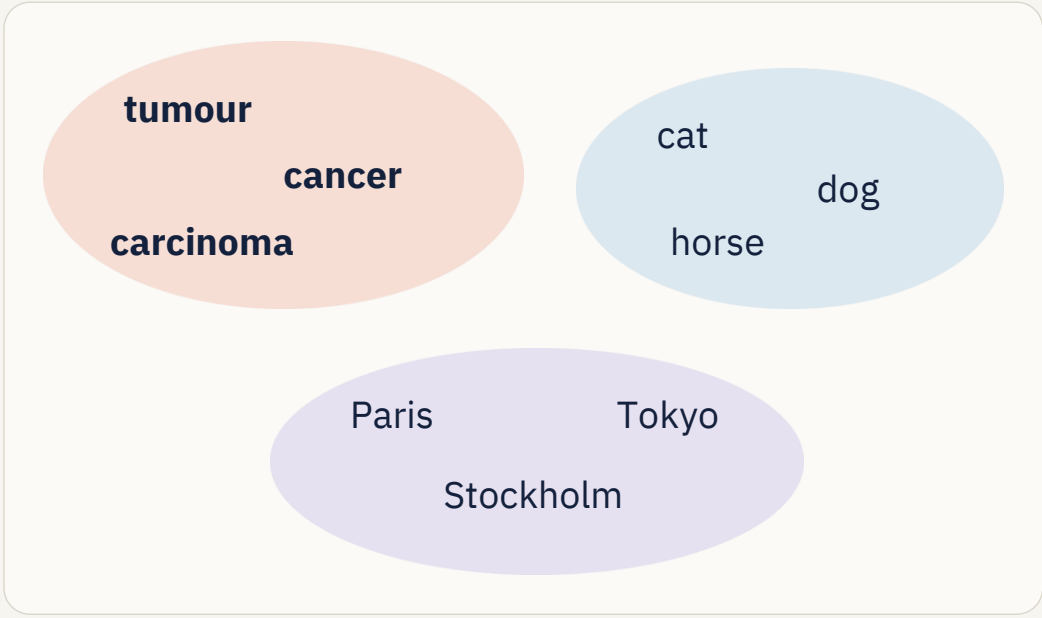
A token is a word or a piece of one. Each gets an ID number (IDs here are illustrative).

2 · Turn each token into a vector

```
"tumour" → [0.12, −0.83, 0.47, … ]
```

This list of thousands of numbers is the token's **embedding** — its meaning as coordinates.

3 · Similar meanings sit close together



A simplified 2-D view; real embeddings have thousands of dimensions.

Attention: which words matter here?

The **animal** didn't cross the street because **it** was too tired

Shading = how much the word **“it”** attends to each other word: **animal** strongest, **tired** medium, **street** weak.

Attention

Every word looks at every other word and decides which ones help explain it.

Transformer

Dozens of attention layers stacked. Processes all words at once — so it scales.

Not just words

Cut an image into patches and the same idea works: Vision Transformers (ViT).

Three stages of training

Stage 1

Pre-training

Read a huge slice of the internet, books and code. Learn to predict the next token.

Result: a “base model” — knows language, but isn't a helpful assistant yet.

Stage 2

Instruction tuning

Train on many examples of good questions and answers written by people.

Result: follows instructions and answers like an assistant.

Stage 3

Human feedback

People rank the model's answers; it learns to prefer the better ones (RLHF).

Result: more helpful, honest and safe.

Analogy: **read the library** → **apprenticeship** → **feedback from a mentor**.



VIDEO BREAK · ABOUT 8 MINUTES

Large Language Models explained briefly

3Blue1Brown, made for the Computer History Museum. A visual, jargon-light recap of everything we just covered.

▶ youtube.com/watch?v=LPZh9BOjkQs

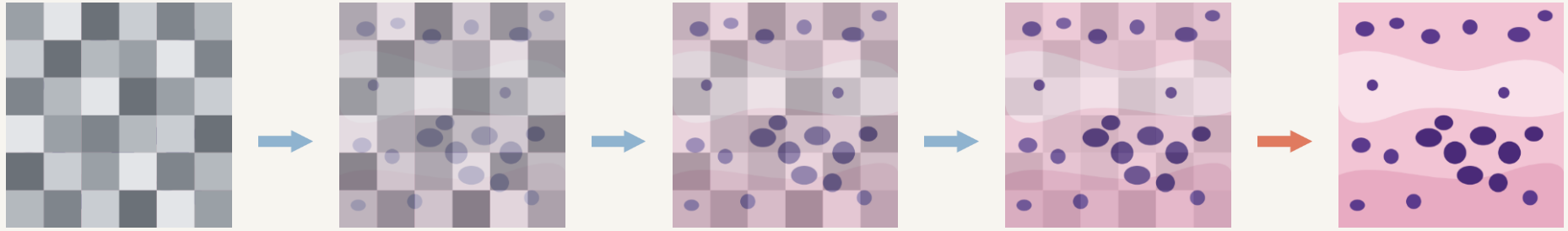
GOING DEEPER AT HOME (OPTIONAL)

Transformers, the tech behind LLMs — 3Blue1Brown
Intro to Large Language Models (1 h talk) — Andrej Karpathy

GENERATING IMAGES

Diffusion: creating by removing noise

Prompt: “H&E-stained breast tissue, high magnification”



Pure noise | Step 250 | Step 500 | Step 750 | **Final image**

Training

Add noise to millions of real images, step by step. Learn to undo each step.

Generating

Start from pure noise; remove it step by step, guided by the text prompt.

IN PRACTICE

Four ways to put an LLM to work

Prompting

Clear instructions, context, examples and the format you want.

“You are a tutor. Explain to a nurse, in 5 bullet points.”

RAG

Retrieval-augmented generation: look up your own documents first, then answer from them.

A chatbot that answers from the hospital's own guidelines

Fine-tuning

Train an existing model a little further on your own examples.

A model that writes in your clinic's report style

Agents

The LLM plans steps and calls tools: search, code, databases.

Pulls the data, runs the analysis, drafts the report

Easier, cheaper → **More effort, more control**

LIMITATIONS

What can go wrong

Hallucinations
Confident but wrong answers; invented references.

Bias
Copies imbalances and stereotypes in its training data.

Privacy
Never paste patient or confidential data into public tools.

Copyright
Open questions about training data and who owns outputs.

Security
Jailbreaks and hidden “prompt injection” instructions.

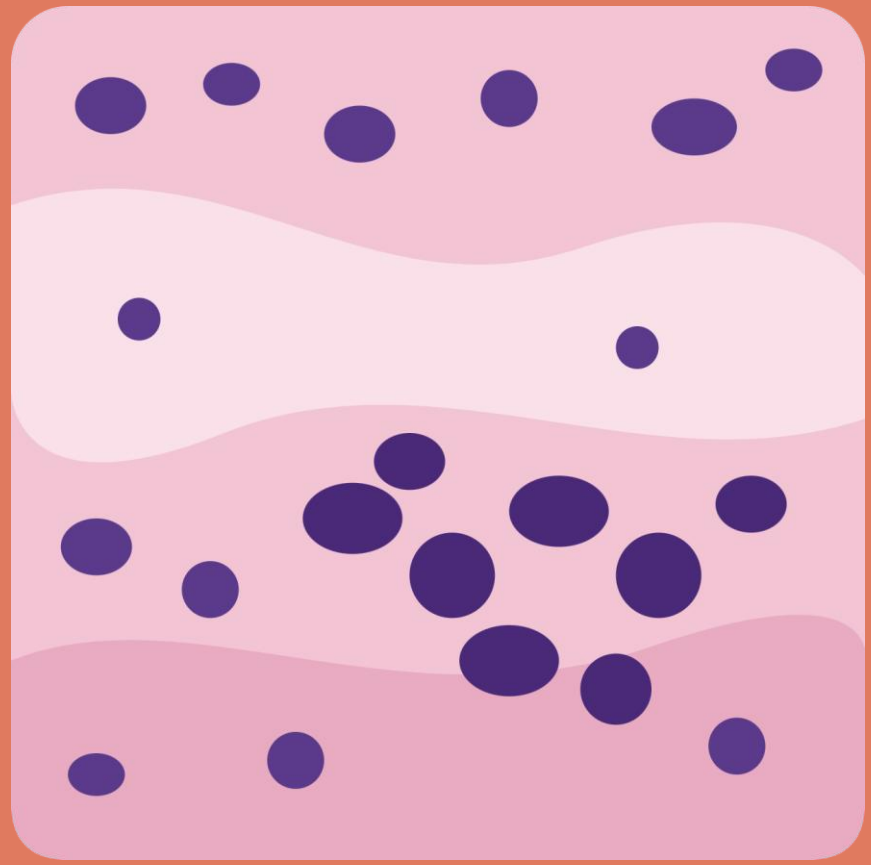
Cost & energy
Large models need a lot of computing power and electricity.

Rule of thumb: treat GenAI output as a **first draft** — and keep a human in the loop.

CASE STUDY ·

Generative AI for cancer image analysis

Breast cancer histopathology: from glass slide to decision support



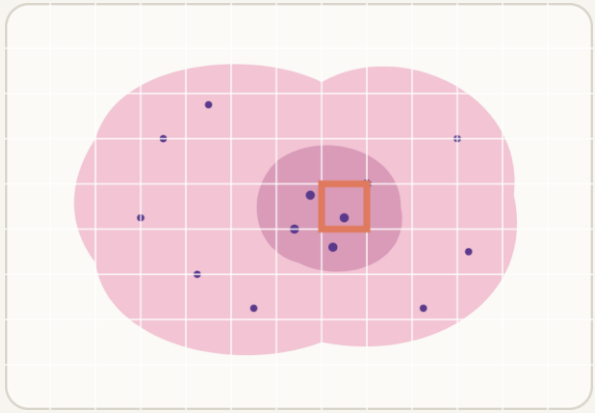
Stylised illustration of H&E-stained tissue — not patient data

CASE STUDY · THE PROBLEM

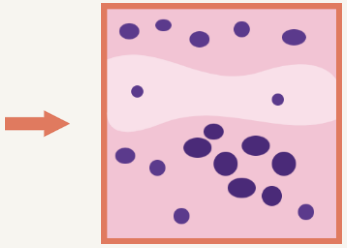
Reading a biopsy is hard, slow work

A pathologist examines stained tissue (H&E) and decides: **benign or malignant?** Which subtype? What grade?

- **Gigapixel images** — one scanned slide can hold billions of pixels
- **Workload** — growing case numbers, too few pathologists
- **Variability** — experts sometimes disagree on borderline cases
- **Few labels** — expert annotations are slow and expensive



Whole-slide image, cut into a grid of tiles



One tile, e.g. 256 × 256 px. A slide yields **thousands** of tiles.

CASE STUDY · THE TOOLBOX

Five ways GenAI can help

1

Foundation models

Pre-trained on millions of unlabelled tissue images; reused for many tasks with few labels.

e.g. UNI, Virchow

2

Image + text models

Classify or search images by describing classes in words — zero-shot.

e.g. CONCH

3

Chat assistants

Ask questions about an image in plain language and get explanations.

e.g. PathChat

4

Synthetic data

Diffusion models or GANs create realistic tiles for rare classes, training and teaching.

Augment small datasets

5

Report drafting

An LLM turns findings into a structured report or a patient-friendly summary.

Pathologist edits & signs

Next: a closer look at **1, 2 and 3**.

CASE STUDY · 1 · FOUNDATION MODELS

From slide to suggestion

1
Whole-slide image
Scan the glass slide

→

2
Tiles
Cut into thousands of patches

→

3 · GenAI
Foundation model
Turns each tile into an embedding

→

4
Small classifier
Combines all tiles (needs few labels)

→

5
Prediction + heatmap
Benign or malignant, subtype — and where

→

6 · GenAI
Draft report
LLM writes it; the pathologist reviews

Why it helps

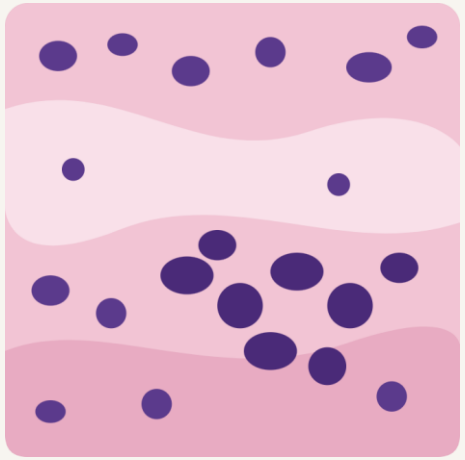
One pre-trained encoder serves many tasks — subtyping, grading, biomarkers — with far fewer labelled slides.

Example: UNI (Nature Medicine, 2024)

Self-supervised on 100 million+ tissue images from 100,000+ slides; outperformed earlier models across 34 tasks.

CASE STUDY · 2 · IMAGE + TEXT

Classify by describing it in words



New image, no task-specific training

SIMILARITY TO EACH WRITTEN DESCRIPTION

“an H&E image of invasive ductal carcinoma”

0.81 · best match

“an H&E image of benign breast tissue”

0.12

“an H&E image of invasive lobular carcinoma”

0.07

Images and text share **one embedding space**, so “closest description” = predicted class. Scores illustrative. Approach: CONCH (Nature Medicine, 2024), trained on 1.17 million image–caption pairs.

CASE STUDY · 3 · CHAT ASSISTANTS

Talking to the image

Illustrative example



What do you see in this breast biopsy region?

Nests and cords of atypical epithelial cells infiltrate the surrounding stroma. The nuclei are enlarged and vary in size and shape. These features suggest **invasive carcinoma**. Grading needs review of the whole slide — please confirm with a pathologist.

PathChat (Nature, 2024)

- ✓ Pathology image encoder (UNI) + a large language model
- ✓ Fine-tuned on 456,000+ image–instruction examples
- ✓ Beat general assistants such as GPT-4V on diagnostic questions in the study
- ✓ Uses: teaching, research, human-in-the-loop support

CASE STUDY · REALITY CHECK

Promising — but not yet autopilot

What the research shows

UNI & CONCH · Nature Medicine, 2024

Large pathology foundation models (image-only and image + text) that beat earlier approaches on many tasks.

PathChat · Nature, 2024

Pathologists preferred its answers over general-purpose chatbots.

Benchmark of 19 models · Nat. Biomed. Eng., 2025

Image + text model CONCH best overall on 31 tasks; Virchow2 a close second.

Before it reaches patients

- Validate on other hospitals, scanners and stains
- Check for bias across patient groups
- Explain decisions — show *where* (heatmaps)
- Regulatory approval (e.g. FDA, EU MDR, EU AI Act)
- **AI suggests, the pathologist decides**

WRAP-UP

Four things to remember

1. GenAI **creates** content. An LLM is a next-word predictor trained at enormous scale.
2. Transformers and **attention** power both language and image models.
3. Start simple: prompting → RAG → fine-tuning → agents. **Always verify.**
4. In pathology, foundation and image + text models cut labelling needs — **humans stay in charge**.

DISCUSSION ·

Over to you

Would you accept an AI “second opinion” on your own biopsy? What would you need to see first?

Who is responsible when an AI-assisted diagnosis is wrong?

Pick a field you know. Where could GenAI help — and what could go wrong?

FURTHER WATCHING & READING

Resources

Videos

Large Language Models explained briefly
3Blue1Brown · ~8 min

Transformers, the tech behind LLMs
3Blue1Brown · ~27 min

Intro to Large Language Models
Andrej Karpathy · ~1 h

How AI image generators work
Computerphile

Explore & read

LLM Visualization — interactive 3-D walk-through

The Illustrated Transformer — Jay Alammar

UNI · CONCH — Nature Medicine, 2024

PathChat — Nature, 2024 (preprint)

Benchmarking pathology foundation models — Nat. Biomed. Eng., 2025

Overview of the Course

ROADMAP

In the course, we cover:

Foundations

01 Overview, history and applications

02 Intelligent agents and knowledge representation

Problem solving

03 Graphs and search

04 Optimization and genetic algorithms

Learning from data

05 Data mining and knowledge discovery

06 Supervised and unsupervised learning

07 Deep learning

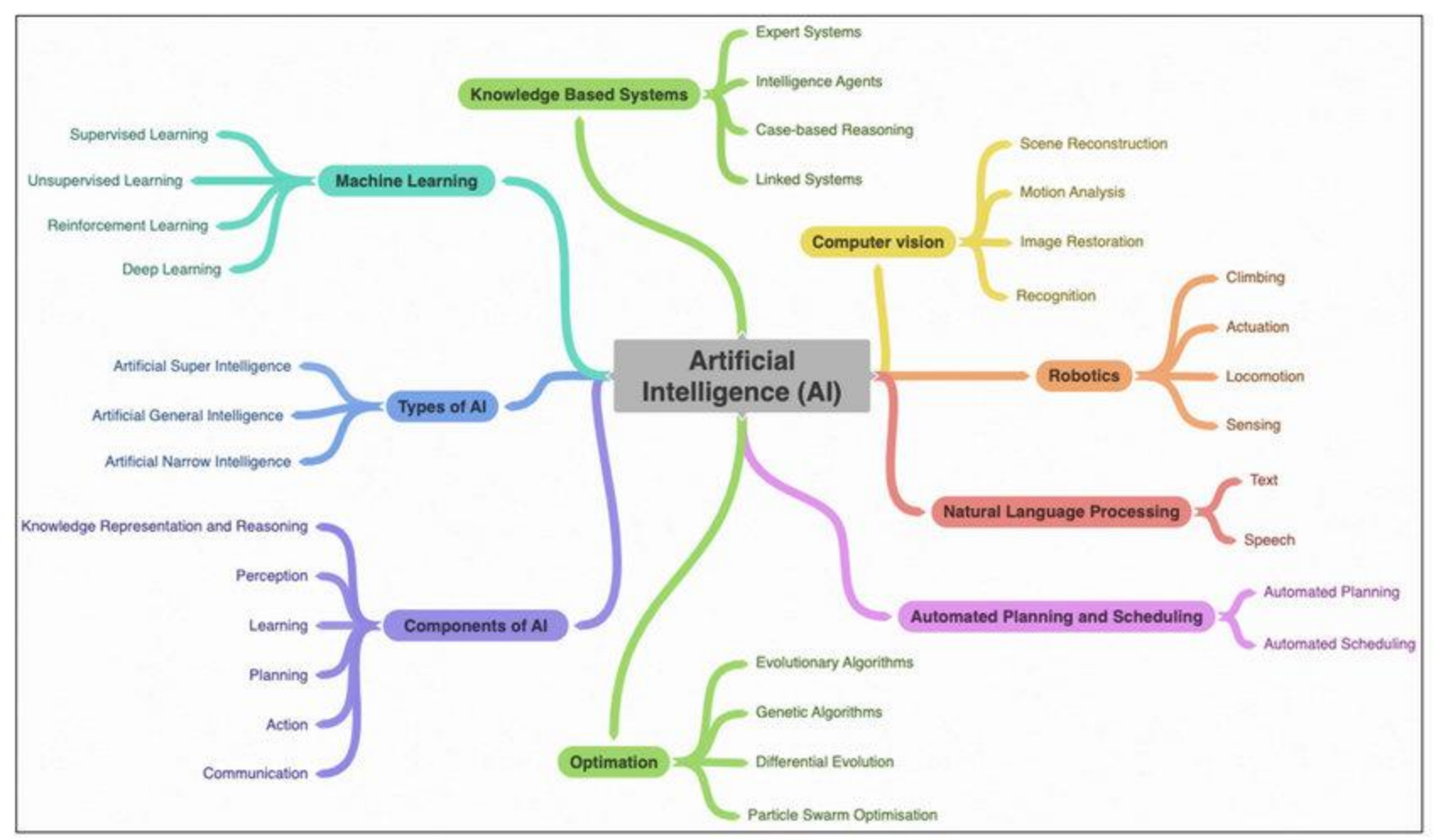
AI in the world

08 Vision and language

09 Ethics, benefits and the future

10 Generative AI and LLMs

What is Artificial Intelligence?


Artificial Intelligence (AI)
Knowledge Based Systems
Expert Systems
Intelligence Agents
Case-based Reasoning
Linked Systems
Machine Learning
Supervised Learning
Unsupervised Learning
Reinforcement Learning
Deep Learning
Types of AI
Artificial Super Intelligence
Artificial General Intelligence
Artificial Narrow Intelligence
Components of AI
Knowledge Representation and Reasoning
Perception
Learning
Planning
Action
Communication
Computer vision
Scene Reconstruction
Motion Analysis
Image Restoration
Recognition
Robotics
Climbing
Actuation
Locomotion
Sensing
Natural Language Processing
Text
Speech
Automated Planning and Scheduling
Automated Planning
Automated Scheduling
Optimation
Evolutionary Algorithms
Genetic Algorithms
Differential Evolution
Particle Swarm Optimisation

Artificial Intelligence

Views of AI fall into four categories (System):

Acting Humanly	Acting Rationally
Thinking Humanly	Thinking Rationally

* Systems that think like humans.
* Systems that act like humans.
* Systems that think rationally.
* Systems that act rationally.

LECTURE 1 · INTRODUCTION

What AI is, and how it got here

1950

Turing asks “Can machines think?” — the Turing test

1956

Dartmouth workshop names the field “artificial intelligence”

1970s–80s

Expert systems boom, then funding “AI winters”

1997

IBM Deep Blue beats chess champion Kasparov

2012

Deep learning breakthrough in image recognition

2022

ChatGPT brings generative AI to everyone

Four ways to define AI

Thinking or acting, like humans or rationally. This course takes the **rational agent** view: a system that acts to achieve the best outcome.

Narrow AI today

Excellent at specific tasks: healthcare, finance, transport, recommendations, virtual assistants. General, human-level AI is still a goal.

Motivation



AI Algorithms are needed for tasks which are too complex for humans.

- Impractical
- Time consuming
- Hard to solve

- The size and complexity of these data sets means that humans are unable to extract useful information from them.

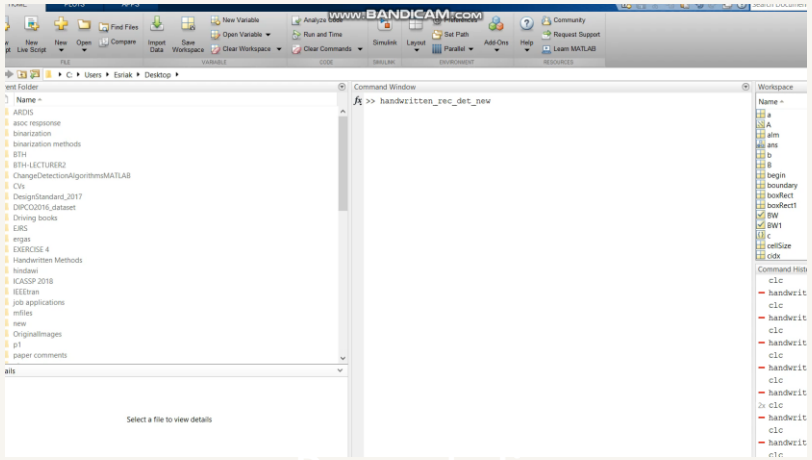
- For a large amount of data, an AI algorithm will work out by exploring that data to predict the result.

Data is everywhere in abundant amounts.

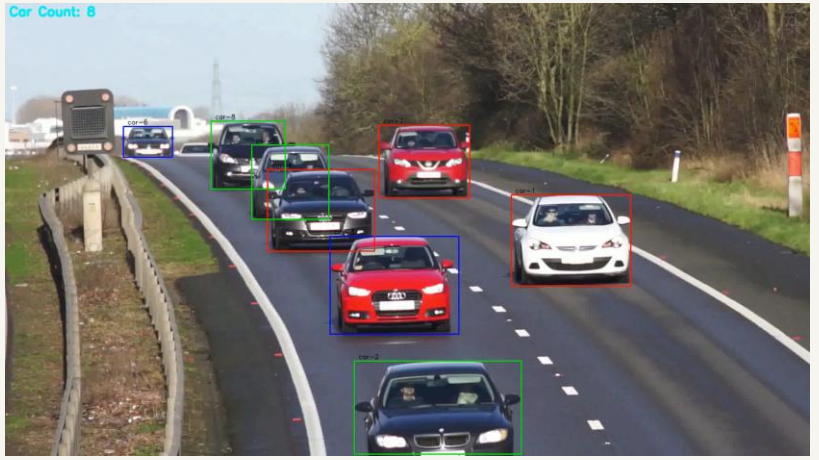
For instance:

- Surveillance cameras continuously capture video,
- Most financial transactions are recorded
- Satellites generate tera-bytes of data every year
- Digitized written texts from our libraries etc.

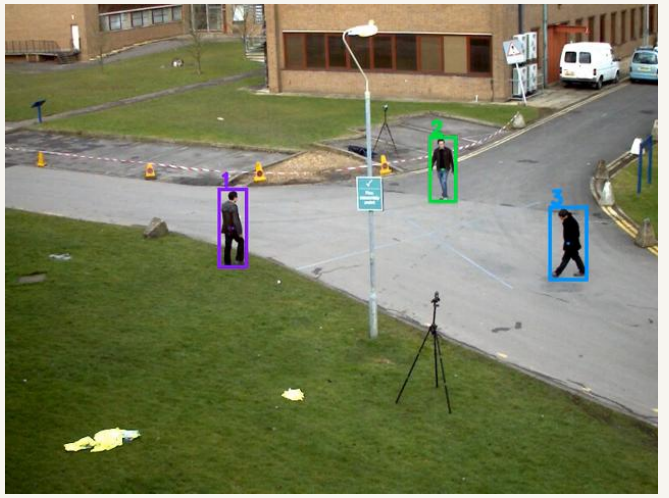
The main purposes of applying AI in human life are to make easier, safer and better.



Demonstration







An agent perceives, decides and acts

Rational agent

Perceives the environment through **sensors**, acts through **actuators**, and chooses the action expected to maximize its performance measure.

PEAS: a self-driving taxi

- **P**erformance: safe, fast, legal trip
- **E**nvironment: roads, traffic, people
- **A**ctuators: steering, brakes, horn
- **S**ensors: cameras, GPS, radar

Knowledge representation

Store facts so the agent can reason: propositional and first-order logic, rules.

Agent types, from simplest to smartest

Simple reflex → Model-based → Goal-based → Utility-based

Problem solving as search through a graph

A problem = **initial state** · **actions** · **transition model** · **goal test** · **path cost**. States are nodes, actions are edges; search finds a path to the goal.

Uninformed (blind) search

No knowledge of where the goal is

- **Breadth-first:** shallowest node first; complete, optimal if all steps cost the same
- **Depth-first:** deepest node first; little memory, may get lost

Informed (heuristic) search

Uses h(n): an estimate of the distance to the goal

- **Greedy best-first:** expands the lowest h(n); fast but not optimal
- **A*:** expands the lowest f(n) = g(n) + h(n), cost so far plus estimate to go
- A* is optimal when h(n) never overestimates (admissible)

Finding the best solution, not just any

1. **Initialize** a random population
2. **Evaluate** each one’s fitness
3. **Select** the fittest as parents
4. **Crossover:** mix parents’ genes
5. **Mutate:** small random changes

Why genetic algorithms?

Inspired by natural evolution. They search huge spaces where no formula exists: scheduling, design, route planning.

Turning raw data into useful knowledge

Selection
pick the relevant data

→

Preprocessing
clean errors, fill gaps

→

Transformation
reduce, encode, scale

→

Data mining
find the patterns

→

Interpretation
evaluate → knowledge

The KDD process: data mining is only one step in a longer pipeline.

Typical tasks

Classification · Clustering · Association rules · Regression · Anomaly detection

Classic examples

Market-basket analysis: customers who buy bread often buy butter. **Decision trees:** readable if–then rules for decisions.

Learning from data, not hand-written rules

Supervised

Learns from labeled examples

- **Classification:** spam or not spam?
- **Regression:** predict a house price
- Decision trees, k-NN, SVM, linear regression

Unsupervised

Finds structure in unlabeled data

- **Clustering:** customer segments
- **Dimensionality reduction:** PCA
- k-means groups points by similarity

Reinforcement

Learns by trial, error and reward

- Game playing (e.g. AlphaGo)
- Robotics and control
- Maximizes long-term reward

Train / test split: judge a model on data it has never seen

Overfitting: memorizes training data, fails on new data

Evaluation: accuracy, precision, recall, error

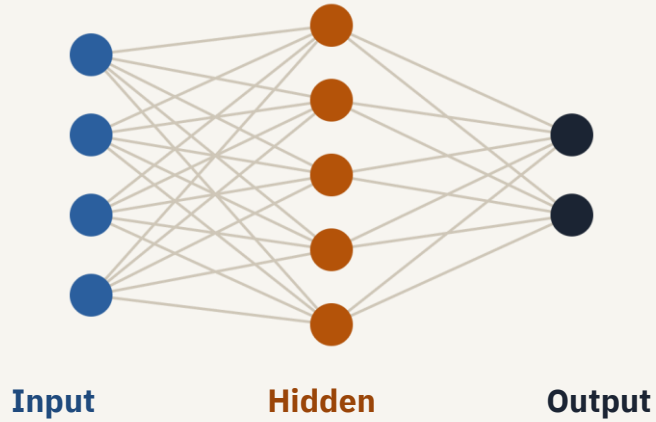
LECTURE 7 · DEEP LEARNING

Neural networks with many layers

A neuron: multiply inputs by weights, add a bias, pass the sum through an activation function.

A network: neurons in layers, input → hidden → output.

“Deep” means many hidden layers, which learn useful features automatically instead of us designing them.



Training

Backpropagation sends the error backwards; gradient descent adjusts the weights.

CNNs

For images: filters detect edges, then shapes, then objects.

RNNs and Transformers

For sequences and language; Transformers power today’s LLMs.

Teaching machines to see and to read

Computer vision

From pixels to meaning

- **Classification:** what is in the image?
- **Object detection:** where is it?
- **Segmentation:** which pixels belong to it?
- CNNs learn edges → shapes → objects
- Uses: medical scans, face unlock, self-driving cars

Natural language processing

From words to meaning

- **Tokenization:** split text into words or pieces
- **Word embeddings:** words as vectors; similar meanings sit close together
- **Language models** predict the next word
- Uses: sentiment analysis, translation, question answering, chatbots

Powerful technology, real responsibilities

Bias and fairness

Models can copy unfair patterns from historical data.

Privacy

Training and surveillance rely on personal data.

Transparency

“Black box” decisions are hard to explain.

Accountability

Who is responsible when AI gets it wrong?

Jobs and economy

Tasks are automated; new skills are needed.

Safety and misuse

Deepfakes, misinformation, cyber-attacks.

Benefits

Faster diagnosis, higher productivity, scientific discovery, accessibility.

The way forward

Regulation (e.g. EU AI Act), human oversight, responsible design.

AI that creates text, images, code and audio

How an LLM is built

1. **Pre-training:** read huge amounts of text and learn to predict the next token (a Transformer)
2. **Fine-tuning:** learn to follow instructions, guided by human feedback (RLHF)
3. **Prompting:** you give instructions and context; it generates a response

Key terms

Token	a piece of a word
Context window	how much text it can consider at once
Hallucination	confident but false output
RAG	retrieve trusted documents, then answer
Multimodal	works with text, images and audio

Use it as a powerful assistant, and always check facts, sources and bias.

In the course:

Assignments and Examination

You had;

- 3 Assignments for DV2618 (Assignment 1, 2 and 3)
- 2 Assignments for DV2659 (Assignment 1 and 3)

Majed is assessing Assignment 1 and **Thi** is assessing Assignment 2
Assignment 3 submission deadline: 18 Oct. 2026 (until 23:59)

- IMPORTANT
- First Resubmission will be available in January, 2026
- Second Resubmission will be available in May, 2026

About Lectures

There are

- 10 lectures
- 3 Labs

Purpose of this course

- Learn how to use the basic search methods;
- Understand the basic methods for problem formulation and knowledge representation;
- Understand the basic idea of automatic reasoning;
- Know some basic concepts related to optimization, data science, prediction, pattern recognition and machine learning.

Recommended research papers

You can find recommended research papers on the course webpage in Canvas.

About Written Exam

It will be on campus

It will be a computer –based exam (Inspera)

About Questions

- ✓No coding
- ✓The questions will be related to the contents taught during the course (e.g., lectures, assignments and labs)
- ✓There can be simple calculations and **you sould bring your own calculator** (bring your own calculator, mobile phones are **NOT** allowed to be used as calculator)
- ✓The exam date will be announced. Please register for thye exam.

Questions?